The Right Grease in the Right Place

Why using one grease for everything is the mistake that destroys components — and the data to prove it

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bicycle-grease-tribology-SKF-Motorex-FinishLine-LiquiMoly-MucOff-professional-workshop-trujillo-peru-bikelab

BikeLab Studio lubrication lineup // SKF · Motorex · Finish Line · Liqui-Moly · Würth · Muc-Off // Every product has one exact purpose. None replaces the other.

Walk into most bike shops in South America and you'll find one can. One. Hardware store multi-purpose grease, car grease, sometimes even petroleum jelly.

They use it on pedals, bearings, seat posts, stems, cassette threads. Everything the same.

That is technically wrong. Not a matter of opinion — a matter of materials engineering.

Tribology — the science that studies friction, wear, and lubrication between surfaces in relative motion — establishes with precision that each mechanical interface has distinct conditions of load, temperature, sliding velocity, and chemical compatibility. Applying the wrong lubricant is not neutral: it accelerates wear, can contaminate brake fluid through migration, and generates incorrect clamping forces in threaded fasteners.

In this photo are the six products I use at BikeLab Studio. Below I explain what each one does, why I selected it over its alternatives, and what workshop data supports those decisions.

MODULE_01 // TRIBOLOGICAL_FUNDAMENTALS

Before discussing products, you need to understand the logic behind them. There are three lubrication regimes that matter on a bicycle:

Hydrodynamic lubrication: A continuous lubricant film completely separates the surfaces. This occurs in bearings rotating at steady speed under moderate load. You need a grease with adequate base viscosity and a thickener that retains the film.

Boundary lubrication: Surfaces partially contact each other — the film is not continuous. This occurs under high load and low speed, as in a bottom bracket during a sprint or a fastener being tightened. EP (Extreme Pressure) additives or solid particles are required to act as a physical cushion.

Mixed lubrication: A combination of both. Most bicycle bearings operate here under variable load and speed conditions.

$$\mu = \frac{F_f}{F_n}$$

Amonton-Coulomb's Law: friction coefficient μ = friction force / normal force. The lubricant reduces F_f. The wrong grease may reduce it insufficiently — or, in cases like

silicone contamination on brake rotors, actually increase it until the braking system fails.

Interface	Tribological regime	Peak load estimate	Lubricant requirement
Hub bearing	Mixed / Hydrodynamic	500–900 N	EP grease, Li or polyurea thickener
Bottom bracket (sprint load)	Boundary	**1,200–1,800 N**	High-load EP grease, AW additives
Stem / seatpost clamp threads	Static contact + micro-vibration	Torque 5–8 Nm	Anti-seize or ceramic paste
Cassette / pedal threads	Galvanic static contact Al–Fe	High load + cyclic moisture	Copper anti-seize (galvanic inhibitor)
Fork / seatpost exposed to rain	Mixed low-speed + water washout	Variable	High water-resistance grease

Source: Stachowiak & Batchelor (2014). Engineering Tribology. 4th ed. Elsevier. / ISO 6743-9 Classification of Lubricating Greases.

Each row in that table is a different product. There is no grease that resolves all those conditions well simultaneously. There is only grease that resolves them all poorly.

MODULE_02 // SKF_LGHP_2-1 // PRECISION_BEARING_GREASE

SKF LGHP 2/1

MINERAL-SYNTHETIC GREASE // POLYUREA THICKENER // HIGH TEMPERATURE

Base viscosity: 100 cSt @ 40°C / Operating range: −20°C to +150°C continuous, peaks to +220°C / NLGI Grade 2 / No lithium thickener.

OPTIMAL_USE: SEALED_PREMIUM_BEARINGS

SKF manufactures the bearings found in most premium hubs that come through this workshop — DT Swiss, Chris King, Hope, Shimano XT/XTR. It's no coincidence I use their own grease to service them. SKF designed LGHP 2 specifically for ball and roller bearings under combined radial and axial loads.

The polyurea thickener, instead of conventional lithium, has a property that matters specifically in Trujillo: **superior resistance to hydrolysis by moisture**. Lithium stearate hydrolyzes with water — polyurea does not. In a coastal city with constant sea fog, this has a real and measurable impact on bearing service life.

Parameter	SKF LGHP 2/1	Generic lithium grease	Practical difference
Max. continuous temperature	**+150°C**	+120°C	+30°C thermal margin
Peak temperature (short-term)	**+220°C**	+160°C	+60°C additional under extreme load
Water washout resistance (ASTM D1264)	**<1% loss**	5-15% loss	5-15x better retention
Estimated service interval	**8,000–12,000 km**	3,000–5,000 km	2–3x longer per service charge
NBR/FKM seal compatibility	**Verified by SKF spec**	Variable / uncertified	No seal swelling risk

Source: SKF Group (2024). "LGHP 2 High Performance Bearing Grease — Product Datasheet PDS LGHP2". / ASTM D1264 Water Washout Characteristics of Lubricating Greases.

> **WORKSHOP_DATA:**
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> When a DT Swiss 240 or Chris King hub arrives with 10,000–14,000 km on it, serviced with LGHP 2, the grease is still visually consistent — no oil separation, no oxidation discoloration. With generic lithium grease, the same hub at 4,000–5,000 km already shows degradation: darkening, consistency loss, early corrosion on the bearing races. I've seen this enough times to have no doubt. The service interval isn't an optimistic interpretation of manufacturer guidelines — it's what the grease condition confirms when it arrives on the bench.

Where I don't use it: on threads. LGHP 2 has no anti-seize additives. Applying it to an aluminum-steel threaded interface under high torque guarantees galvanic corrosion at medium term. That's where a different product comes in.

MODULE_03 // MOTOREX_FETT_2000 // THE_WORKHORSE_GREASE

MOTOREX FETT 2000

HIGH PERFORMANCE MULTI-PURPOSE GREASE // LITHIUM THICKENER // NLGI GRADE 2

Base viscosity: 160 cSt @ 40°C / Range: −30°C to +130°C / EP + AW (Anti-Wear) additives / 850g.

OPTIMAL_USE: BOTTOM_BRACKET · HEADSET · MID-RANGE_HUBS

Motorex is Swiss, founded in 1917. They are not a cycling brand — they are an industrial lubricant company with a dedicated bicycle line. That distinction matters: their formulation base comes from industrial engineering, not sports marketing.

Fett 2000 has a significantly higher base viscosity than most bicycle greases: 160 cSt versus the typical 68–100 cSt. Higher base viscosity means a thicker film under compression. This makes it ideal for low-speed, high-load interfaces — exactly what a bottom bracket and headset are.

$$\eta = \frac{\tau}{\dot{\gamma}}$$

Dynamic viscosity η = shear stress τ / shear rate γ̇. Higher η means greater resistance to film breakdown under load. Fett 2000 maintains a continuous film where lighter greases fail at the bearing interface during sprint efforts.

Application	Fett 2000	Estimated interval	Workshop observation
Threaded BB (BSA)	**Optimal**	6,000–8,000 km	No creaks, no thread corrosion
Integrated headset 1-1/8"	**Optimal**	8,000–10,000 km	Smooth action through full interval
Pedal axle	**Optimal**	4,000–6,000 km	Resolves the "click" misattributed to BB
Mid-range hub bearings	Adequate	5,000–7,000 km	Correct. For premium components, SKF LGHP.

Source: Motorex AG (2024). "Fett 2000 Technical Data Sheet — Ref. 300743". Motorex Oil of Switzerland. / Intervals: own field estimates verified at BikeLab Studio 2023–2026.

WORKSHOP_DATA:

The bottom bracket creak is the most frequent complaint in bikes that have been through another shop. In a high percentage of cases that come through here, the cause is not the BB — it's the pedal axle thread, dry or with degraded grease. I apply Fett 2000 to pedal threads at the correct torque (40 Nm left side; reverse thread) and the noise disappears. No new parts. This isn't brilliant diagnosis — it's basic maintenance applied correctly.

MODULE_04 // FINISH_LINE_CERAMIC_GREASE // WHERE_THE_FILM_BREAKS

FINISH LINE CERAMIC GREASE

SYNTHETIC PAO GREASE // SUBMICRON CERAMIC PARTICLES // NLGI GRADE 1.5

Base: PAO (Polyalphaolefin) synthetic / Particles: TiO_2 + Al_2O_3 submicron / Range: −40°C to +200°C / Compatible with all bicycle components.

OPTIMAL_USE: CERAMIC_BEARINGS · CARBON_CONTACT · SPD_CLEATS

Ceramic greases operate on a different principle than conventional ones. The ceramic particles — titanium oxide and alumina in this case — act as microscopic bearings between metal surfaces. When the oil film breaks under extreme load (boundary lubrication regime), the solid particles prevent direct metal-to-metal contact. This isn't marketing copy — it's basic contact tribology.

The PAO (polyalphaolefin) base provides two advantages over mineral bases: **more stable viscosity index across temperature**, and **superior oxidation resistance** without precipitating paraffins. This matters especially in ceramic bearings, where aggressive chemical additives in conventional greases can attack the retention rings.

Property	Mineral NLGI 2 grease	Finish Line Ceramic (PAO)	Practical difference
Viscosity Index (VI)	95–110	**140–165**	Less thermal film variation
Boundary lubrication protection	Fluid film only	**Film + solid ceramic barrier**	Protection when film breaks under load

Property	Mineral NLGI 2 grease	Finish Line Ceramic (PAO)	Practical difference
Carbon fiber compatibility	**Variable — possible aggressive solvents**	**Verified — non-aggressive with epoxy resin**	No resin degradation on contact
Minimum operating temperature	-20°C	**-40°C**	Relevant for Andean cycling above 4,000m

Source: Finish Line Technologies (2024). "Ceramic Grease Technical Data Sheet". / Rudnick, L.R. (Ed.) (2013). Synthetics, Mineral Oils, and Bio-Based Lubricants. CRC Press.

> **WORKSHOP_DATA:**
>
> In hybrid ceramic bearings (ceramic balls, steel races), I've measured rolling resistance differences of 4–7% compared to the same bearing regreased with conventional lithium grease. The PAO base has a slightly lower friction coefficient, and the ceramic particles don't generate the same drag as conventional grease solid thickeners at low loads. Not huge numbers — but real, measurable watts.

MODULE_05 // LIQUI-MOLY_COPPER_PASTE // WHERE_GREASE_DOESN'T_BELONG

LIQUI-MOLY KUPFERPASTE (COPPER PASTE)

COPPER ANTI-SEIZE COMPOUND // NOT A BEARING GREASE // THREADED AND GALVANIC INTERFACES

Copper particles in suspension / Controlled friction coefficient: μ = 0.13 / Temperature: up to +1,100°C / Galvanic corrosion inhibitor (Al–Fe).

CORRECT_USE: THREADS · BIMETALLIC_INTERFACES · CABLE_STOPS

This is probably the most misunderstood product here. Copper paste is not a grease for rotating parts — it's an anti-seize compound. Its function is to prevent two dissimilar metals in contact, under load and moisture, from corroding to the point where disassembly becomes impossible.

On a bicycle, galvanic corrosion between aluminum and steel is a real and serious problem. Steel bolts in aluminum stems, pedal threads (steel) in aluminum cranks, brake mounts (titanium or steel) in steel frames — all are bimetallic interfaces under cyclic load and moisture exposure.

$$\Delta V_{galvanic} = E_{cathode} - E_{anode}$$

Electrochemical potential difference: Al = −1.66V vs Fe/steel = −0.44V → ΔV = 1.22V. An active, corrosive galvanic couple. Copper (E = −0.34V) interposes an intermediate potential that suppresses galvanic current without electrically isolating the parts.

CRITICAL_TORQUE_WARNING:

Copper paste modifies the friction coefficient in threads — μ shifts from ~0.20 (dry) to ~0.13. That changes the torque-to-preload conversion: if you apply paste and then torque to the dry spec, you are preloading the bolt 35–50% beyond specification. On carbon fiber stems and handlebars, this can fracture the insert. Most manufacturers (Trek, Specialized, Canyon) specify dry torque values. Always verify before applying.

Interface	Use Copper Paste?	Technical reason
Pedal thread (steel in aluminum crank)	**YES**	Active galvanic couple + high load cycles
Cassette lockring thread (Al/steel)	**YES (minimal dose)**	Prevents seize, ensures clean removal
Cable stops on steel frame	**YES**	Prevents ferrous seize at weld points
Stem bolts on carbon fiber frame	**NO (normally)**	Modifies torque — risk of insert fracture
Rotating bearings	**NEVER**	Copper particles are abrasive on steel races at speed. Destroys the bearing.

Source: Liqui-Moly GmbH (2024). "Kupferpaste Technical Data Sheet Ref. 3080". / Bickford, J.H. (2007). Introduction to the Design and Behavior of Bolted Joints. 4th ed. CRC Press. / VDI 2230 Systematic Calculation of Bolted Joints.

FINISH LINE PREMIUM BICYCLE GREASE

BICYCLE GREASE / CALCIUM SULFONATE THICKENER / HIGH WATER RESISTANCE / NLGI GRADE 2

Highly refined mineral base / Thickener: calcium sulfonate / Certified waterproof / 450g.

CORRECT_USE: SEATPOST · WET_FORK · EXPOSED_HEADSET_CUPS

Calcium sulfonate thickener has a property most lithium greases can't match: adhesion to metal in the presence of water. When a lithium-greased interface gets wet, water can displace the grease from metal surfaces. With calcium sulfonate, the grease bonds to the substrate and water rolls off without penetrating the lubricant film.

This is not the most technically sophisticated product in the lineup. No ceramic particles, no synthetic base. But for what it does, it does it better than the alternatives: **it maintains a lubricant film in wet conditions where most greases fail**.

Property	Typical lithium thickener	Calcium Sulfonate (FL Premium)
Water washout resistance (ASTM D1264)	3-8% loss	**<1% loss**
Drop point	170-200°C	**>300°C (no dropping)**
Rust inhibition (ASTM D1743)	Passes (variable)	**Passes (consistent)**
Oil separation under temperature	Moderate	**Very low**

Source: Finish Line Technologies (2024). "Premium Bicycle Grease Technical Specification". / ASTM D1264 / ASTM D1743 Standard Test Methods for Lubricating Greases.

MODULE_07 // MUC-OFF_SILICON_SHINE // WHAT_ISN'T_A_LUBRICANT

MUC-OFF SILICON SHINE

SILICONE PROTECTIVE SPRAY // NOT A LUBRICANT FOR ROTATING COMPONENTS

Polydimethylsiloxane (PDMS silicone) in solvent base / UV protection / Water and dirt repellent / 500ml.

CORRECT_USE: FRAME · PLASTICS · EXTERNAL_CABLE_OUTERS · FINISH

Silicon Shine is not a lubricant in the tribological sense. It is a surface protector. The polydimethylsiloxane (PDMS) forms a non-polar film that repels water and reduces adhesion of dust and mud. I use it for final finishing after a full service: on a clean frame it protects aluminum anodizing from UV oxidation and shields painted frame decals. On external cable outers, it reduces static friction with the housing.

CRITICAL_BRAKE_CONTAMINATION:

Silicone contaminates brake rotors. A spray of Silicon Shine near the rotors deposits an invisible film that dramatically reduces the friction coefficient between pad and rotor. I have received bikes with unexplained fade on brand new brakes — in several cases the cause was silicone. Contaminated pads do not recover with cleaning: they must be replaced. Silicone also does not belong on the chain, bearings, or any interface requiring traction. It is exclusively for external finishing surfaces.

MODULE_08 // WHAT'S_MISSING_FROM_THE_SOUTH_AMERICAN_MARKET

There are products I would use regularly that are not reliably available in Peru or broader South American distribution. I mention them because doing this work properly means knowing what exists beyond what's on the shelf.

PRODUCTS_ON_WATCHLIST // NOT CONSISTENTLY AVAILABLE IN PERU (Feb. 2026):

Dumonde Tech Original Bicycle Grease: Niche synthetic formulation, widely used in cyclocross and XC racing in the US. Very low base viscosity for high-RPM bearings. Field references from WorldTour mechanics are convincing enough to keep it on my radar. Not consistently available here.

Phil Wood Waterproof Grease: An industry classic. Lithium thickener with documented anti-corrosion additives tested across decades on chromoly steel hubs. Sporadically available in Lima. In Trujillo, practically impossible to source.

Krytox GPL 206 (DuPont/Chemours): PFPE (perfluoropolyether) grease — inert to everything. Originally developed for aerospace applications.

> Incompatible with nothing, durable under conditions that would destroy any conventional grease. For bearings in extreme Andean conditions — mud, severe cold, constant moisture — it would be ideal. Price and availability in Peru: essentially zero.
>
> **CeramicSpeed UFO Drip Grease:** Extremely expensive. Probably unjustifiable for the current South American market. The focus on reducing the Coefficient of Rolling Resistance (CRR) with amorphous diamond particles is technically interesting. Noted for the record, not as a recommendation.

MODULE_09 // APPLICATION_MAP // OPERATIONAL_SUMMARY

Bicycle interface	Correct product	Estimated interval	Most common error elsewhere
Premium hub bearings	**SKF LGHP 2/1**	8,000–12,000 km	**Generic lithium grease → service at 3,000 km**
Bottom bracket / headset	**Motorex Fett 2000**	6,000–8,000 km	**Same hub grease, or nothing**
Ceramic bearings / carbon contact	**Finish Line Ceramic**	6,000–10,000 km	**Mineral grease — may degrade epoxy resin**
Pedal / cassette threads, cable stops	**Liqui-Moly Copper Paste**	Every disassembly	**Grease or nothing → guaranteed seize in 2 seasons**
Seatpost / fork exposed to rain	**Finish Line Premium**	4,000–6,000 km	**Any grease → washes out with first rain**
Frame / cable outers / plastics	**Muc-Off Silicon Shine**	After every wash	**Nothing, or applied near rotors → contamination**

Intervals based on direct workshop observation, BikeLab Studio 2023–2026. Over 200 documented services under Trujillo conditions: coastal, moderate humidity, no extreme mud exposure.

CONCLUSIONS // APPLIED_TRIBOLOGY

A single grease cannot resolve all the tribological regimes of a bicycle. That is not an opinion — it is contact mechanics. The parameters of base viscosity, thickener type, EP additives, chemical compatibility and operating temperature cannot be unified by any generic product. All a "one for everything" product achieves is to be mediocre at everything.

What validates this selection is not theory — it's the condition of components when they arrive after two or three seasons. Premium hubs maintained with LGHP 2 arrive with visually intact grease at 12,000 km. Bottom brackets treated with Fett 2000 on their threaded interfaces don't develop the creak clients attribute to the BB. Pedal threads treated with Copper Paste come apart with a normal wrench after three years. That is what validates the selection — not the price of the products.

The cost of the six products in that photo significantly exceeds the cost of one generic can. The difference is that each product does one thing exactly right — and that translates into components that last as long as they technically should.

The right grease in the right place is not perfectionism. It is the difference between maintenance and slow, controlled damage.

PRACTICAL APPLICATIONS

Operational implementation of these principles: lubricant/grease selection by application point (lubricantes-grasas), washing protocol without bearing contamination (lavar-bicicleta), drivetrain maintenance and lubricant film control (cuidado-transmision), chain wear measurement and replacement thresholds (cambio-cadena).

[CLUSTER_DATA_LINKS] // DERIVED_PRACTICAL_GUIDES

→ Applied Tribology: Chain Lube vs Bearing Grease

→ The Mistake of Washing with Pressure Water

→ Drivetrain Care: Preventing Premature Wear

→ Chain Replacement: Thresholds 0.5% / 0.75% / 1%

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